S02. Bandgap optimization in a bounded wave-operator family

Pure Thought research portfolio

Revised 17 September 2026

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Source: synthesis challenge 12; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

Some repeating structures prevent waves in certain frequency ranges from traveling through their interior. The challenge is to choose a structure that makes this forbidden range as large as possible within stated material and geometric limits. A gap must survive every propagation direction, including directions missed by a convenient plotting path.

Key terms

- **Wave operator** — The mathematical rule governing the waves in the chosen physical model.
- **Periodic medium** — A material pattern repeated through space.
- **Unit cell** — One repeating piece of that pattern.
- **Bloch wave** — A wave adapted to a periodic medium, labeled by a wave vector.
- **Brillouin zone** — The set of inequivalent wave vectors needed to describe a periodic medium.
- **Band** — A family of allowed wave frequencies or corresponding operator eigenvalues.
- **Complete bandgap** — A frequency interval containing no bulk wave modes for any wave vector in the specified model.
- **Contrast** — The permitted ratio or variation between material coefficients.
- **Global optimum** — The best value across the entire declared design family, rather than just nearby designs.

Why this matters

Controlling which waves propagate supports ideas for sound isolation, vibration management and optical filtering. The same mathematical challenge arises across these applications: a promising numerical design needs reliable evidence that its gap exists everywhere and that any optimality claim covers the stated design space.

What progress would mean

A successful result would certify a nonzero gap, bound the best achievable objective, or establish an optimum within the bounded family. A scalar-wave benchmark would test the method; applying it to electromagnetic devices requires the appropriate Maxwell model and physical constraints.

Revised core question

Within a fixed finite family of periodic media, what lower bound on a complete bandgap and what upper bound on the best possible gap can be certified?

Precise scope, target and outputs

Choose either a scalar acoustic operator or a full Maxwell operator, never an unspecified mixture. Initially choose a periodic scalar elliptic operator $-\nabla\cdot a(r)\nabla$ with $0<a_min\leq a(r)\leq a_max$, a fixed unit cell and a finite geometry parameter box. For adjacent Bloch eigenvalues define the complete gap $g=\min_k\ \lambda_\{n+1\}(k)-\max_k\ \lambda_n(k)$, with a declared normalized objective. A grid-local separation is not a complete gap. Use validated eigenvalue bounds plus k-space variation estimates to cover the Brillouin zone. Branch-and-bound over parameters yields a candidate lower bound and an optimization upper bound, keeping the inequality direction explicit.

Detailed critique of the synthesis framing

The old description mixes photonic and phononic operators, index contrast and generic gap optimization without fixing constitutive laws, geometry regularity, or constraints. "Lowest possible contrast" and "largest possible gap" are two objectives needing an ordering or Pareto definition. A converged plane-wave plot does not establish a complete gap, and a locally optimized geometry does not prove global optimality. This is distinct from PRD 11's topological invariant: a wide ordinary bandgap can be topologically trivial.

Validation and milestones

Baseline: one certified complete gap for a supplied geometry. Strong: bound the optimum to a declared tolerance over the entire compact parameter box, or report unresolved boxes. Validate a homogeneous medium and a simple one-dimensional layered case before multidimensional optimization.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Displayed formulation and claim boundary

Supplemental specification.

$$\lambda_- = \max_k \lambda_n(k), \qquad \lambda_+ = \min_k \lambda_{n+1}(k), \qquad g = \lambda_+ - \lambda_-.$$

A possible normalized objective is $G = 2g/(\lambda_+ + \lambda_-)$ when the denominator is positive; frequency-gap objectives must instead use $\omega_j = \sqrt{\lambda_j}$ consistently.

Use a fixed self-adjoint wave model, bounded coefficients and geometry class, with a positive denominator. A complete gap needs $\lambda_+ > \lambda_-$ across the entire Brillouin zone and all required polarizations. A certified candidate lower bound does not establish global optimality; that needs coverage of the full design class.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.